MIL-P-46111C(MR)
28 September 1978
SUPERSEDING
MIL-P-46111B(MR)
29 November 1973

MILITARY SPECIFICATION

PLASTIC FOAM, POLYURETHANE (FOR USE IN AIRCRAFT)

This specification is approved for use by the Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of polyurethane foam of 1.5 pounds per cubic foot maximum density, and fire retardant properties.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
- PPP-B-585 - Boxes, Wood, Wirebound
- PPP-B-601 - Boxes, Wood, Cleated-Plywood
- PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY
- MIL-P-116 - Preservation-packaging, Methods of
- MIL-H-5606 - Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
- MIL-T-5624 - Turbine Fuel, Aviation, JP-4 and JP-5 Grades
- MIL-T-6396 - Tank, Fuel Oil, Water-Alcohol, Coolant Fluid, Aircraft, Non-Self-Sealing, Removable, Internal
- MIL-L-7808 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
- MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water Vaporproof or Waterproof, Flexible
- MIL-L-23699 - Lubricating Oil, Aircraft Turbine Engines, Synthetic Base
- MIL-H-83282 - Hydraulic Fluid

FSC 9330

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Army Materials and Mechanics Research Center, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS:

D 883 - Nomenclature Relating to Plastics
D 1564 - Testing Slab Flexible Urethane Foam
E 162 - Surface Flammability of Materials Using a Radiant Heat Energy Source
G 21 - Determining Resistance of Synthetic Polymeric Materials to Fungi

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The plastic foam shall consist of a flexible polyurethane material which has had a fire retardant additive to permit passage of the firing test (see 3.2). Reclaimed polyurethane foam is permitted when it meets all requirements specified herein and the molecular weight is essentially the same as that of the unused polymeric material specified.

3.2 Property values. The foam material shall conform to the property values specified in table I, when tested as specified in the applicable procedure of 4.3.

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Table I. Property values

Property	Values
Density, kilograms per cubic metre (pounds per cubic foot) at 23° ± 1°C (73.4° ± 1.8°F), max.	24 (1.5)
Flammability (flame spread index), max.	75
Firing test, flame extinguishing within	100mm[1/] (4 in.)
Tensile strength, KPa[1/] (psi), min.	69 (10)
Elongation, percent, min.	200
Tear resistance, grams per millimetre (pounds per inch) thickness, min.	36 (2)
Compression set, percent, max.	10
Compression set after autoclaving, percent, max.	15
Indentation load deflection, kilograms per 320 square centimetres (pounds per 50 square inches), min.	
At 25 percent deflection	7.3 (16)
At 65 percent deflection	14 (30)
Compression load deflection after autoclaving, based on load deflection before autoclaving, percent change, max.	±20

[1/] KPa = kilopascal, mm = millimetre

3.3 Resistance to fluids. The foam material shall show no visual evidence of softening or degradation when immersed for 48 ± 1 hour at 23° ± 1°C (73.4° ± 1.8°F) in the fluids specified. The fluids are covered by MIL-H-5606, MIL-T-5624, MIL-L-7808, MIL-L-23699, and MIL-H-83282. Removal of colored dye additive shall not constitute failure.

3.4 Fungus resistance. The foam material shall be fungus resistant as shown by 0 percent growth when examined visually.

3.5 Color. Unless otherwise specified by the procuring activity, color is optional with the supplier except that it shall be soluble in the types of jet fuel specified in MIL-T-5624.

3.6 Dimensions and tolerances. The length, width, thickness, and tolerances for each dimension shall be as specified by the procuring activity (see 6.2).

3.7 Uniformity. The foam material and the color shall be uniform.

3.8 Workmanship. The foam shall be clean and contain no inclusions, large voids, loose particles, or accumulation of unexpanded resin. (These defects may be defined in accordance with ASTM D 883, as applicable.)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Sampling for inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type submitted for inspection at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced, subsidiary specifications and standards to the extent specified. In the event of conflict, this specification shall govern. A suppliers certificate of compliance with 3.1 shall be furnished.

4.2.2 Inspection of the foam.

4.2.2.1 Examination of the foam. Examination shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of packages of plastic foam as applicable, for examination in 4.2.2.1.1, 4.2.2.1.2, 4.2.2.1.3, 4.2.2.1.4 and in units of shipping containers for examination under 4.2.2.1.5.

4.2.2.1.1 Examination of the foam for defects in appearance, construction and workmanship. The sample unit for this examination, specified in table II, shall be one piece of foam one foot in length and one foot in width, as applicable. Not more than five sample units shall be taken from any one package of foam.

Table II. Examination for defects in appearance, uniformity, construction and workmanship

Examine	Defect
Appearance and construction	Color not soluble as specified in 3.5 Color not uniform Foam not uniform
Workmanship	Not clean, presence of inclusions, large voids, loose particles, or accumalation of enexpanded resins.

4.2.2.1.2 Examination of the foam for dimensional defects. The sample unit for this examination specified in talbe III shall be one piece of foam.

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Table III. Examination for dimensional defects

Examine	Defect
Length and width	Varies by more than length and width tolerances specified by the procuring activity.
Thickness	Varies by more than the tolerance specified by the procuring activity.

4.2.2.1.3 Examination of the foam for defects in assembly. The sample unit for this examination, specified in table IV, shall be one piece of foam one foot in length and one foot in width, as applicable.

Table IV. Examination for defects in assembly

Examine	Defect
Assembly of foam	Not evenly stacked

4.2.2.1.4 Examination of the foam for defects in the count per package. The sample unit for this examination specified in table V, shall be one package of foam as applicable.

Table V. Examination for defects in the count per package

Examine	Defect
Foam	Average count per package of foam less than specified

4.2.2.1.5 Examination of the packaging requirements. An examination shall be made in accordance with table VI to determine that packaging and marking comply with the requirements of section 5. The sample unit for this examination shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

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Table VI. Examination of packaging

Examine	Defects
Preservation	Not level specified; not in accordance with contract requirements.
	Packaging material not as specified; closures not made by specified or required methods or materials. Unit quantities not as specified.
Packing	Not level specified; not in accordance with contract requirements.
	Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability.
	Inadequate application of components such as: Incomplete closure of case liners; container flaps, loose or inadequate strappings, bulged or distorted containers.
Count	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings omitted, illegible, incorrect, incomplete, or improper size, location, sequence, method or application, or not in accordance with contract requirements.

4.2.2.1.6 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptable quality levels (AQLs) expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	I	1.5
4.2.2.1.2	S-3	2.5
4.2.2.1.3	S-2	2.5
4.2.2.1.4	S-2	---
4.2.2.1.5	S-2	4.0

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4.2.3 Testing. The foam shall be tested for the applicable characteristics listed in section 3, in accordance with the test methods specified herein for each lot submitted for inspection. Sampling shall be in accordance with MIL-STD-105 and the lot size for the purpose of determining sample size for testing shall be expressed in units of 4.5 kilograms (10 pounds) of material. The inspection level shall be S-1. The sample size (number of units of product drawn from a lot) shall be in accordance with the single sampling plan for normal inspection and each unit shall consist of sufficient material to prepare all specimens required for testing. Each unit of product shall be tested and must meet all applicable requirements of section 3. Test results for each characteristic in each unit of product shall be the averaged result from the specimens tested for that characteristic.

4.2.3.1 Classification of tests. All tests shall be classified as follows:

(a) Lot acceptance tests (see 4.2.3.2).
(b) Periodic lot check test (see 4.2.3.3).

4.2.3.2 Lot acceptance tests. Lot acceptance tests shall be made on each lot of material and shall be the basis for acceptance or rejection of the lot, except where the periodic lot check test is required. The lot acceptance tests shall consist of all tests except the firing test (see 4.3.5).

4.2.3.3 Periodic lot check test. The periodic lot check test shall be made on the first lot of material furnished under this specification, and on any subsequent lot specified by the procuring agency (see 6.2). The periodic lot check test shall consist of the firing test (see 4.3.5). When the periodic lot check test is made, it shall be considered along with results of the lot acceptance tests in determining acceptance or rejection of the lot.

4.3 Test methods.

4.3.1 Test specimens. Test specimens shall be prepared by cutting specimens from the foam sample unit.

4.3.2 Conditioning and testing. (not applicable to the flammability and firing test). Test specimens shall be conditioned in accordance with ASTM D 1564. Testing shall be at 23° $\pm$ 1.1°C (73.4° $\pm$ 2°F) and 50 $\pm$ 2 percent relative humidity.

4.3.3 Density. Two specimens shall be tested in accordance with ASTM D 1564, using the dimensions specified in this method.

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4.3.4 Flammability. Four specimens shall be tested in accordance with ASTM E 162. Length and width of test specimens shall be as specified in ASTM E 162 and thickness shall be that at which the foam material is to be supplied to the procuring activity. Conditioning and testing of specimens shall be as specified in ASTM E 162. The flame spread index for each set of four specimens shall be determined.

4.3.5 Firing test. The standard gunfire test structure specified in figure 4 of MIL-T-6396 shall be used. Three specimens, each between 76 and 82 millimetres (3 and 3-1/4 inches) thick, shall be cut to the length and width of the function plate and placed between the function plate and the standard bladder type fuel cell of the gunfire test structure. One foam face shall be in contact with the function plate and the other in contact with the fuel cell which shall be three quarters full of JP4 fuel conforming to MIL-T-5624. The function plate shall be 6.4 millimetre (1/4 inch) nominal thickness of 2024 T3 aluminum alloy. A firing test shall be made for each of the three specimens using caliber 0.50 API M8 ammunition with a striking velocity equal to or greater than 750 meters (2500 feet) per second at 0° obliquity to the function shall show flame extinguishing within 100 millimetres (4 inches) of the origin in all directions.

4.3.6 Tensile strength and elongation. Three specimens shall be tested in accordance with ASTM D 1564 using the dimensions specified in this method.

4.3.7 Tear resistance. Three specimens shall be tested in accordance with ASTM D 1564 using the dimensions specified in this method.

4.3.8 Compression set. Three specimens shall be tested in accordance with ASTM D 1564, method B, using the dimensions specified in this method. Deflection shall be 50 $\pm$ 1 percent of the thickness. Compression set shall be determined as a percentage of the original thickness.

4.3.9 Compression set after autoclaving. Three specimens, with dimensions as specified in 4.3.8, shall be subjected to condition B of the steam autoclave test specified in ASTM D 1564. Following steam autoclaving, the specimens shall be tested for compression set in accordance with 4.3.8.

4.3.10 Indentation load deflection. Three specimens shall be tested in accordance with ASTM D 1564, using method A. Dimensions for each specimen shall be 380 by 380 by 100 millimetres (15 by 15 by 4 inches) with tolerances of -0, +3 millimetres (-0, +1/8 inch).

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4.3.11 Percent change in compression load deflection after autoclaving. All testing shall be in accordance with ASTM D 1564. Initially, three specimens shall be tested for compression load deflection. Dimensions for each specimen shall be 50 by 50 by 25 millimetres (2 by 2 by 1 inch) with tolerances of -0, +3 millimetres (-0, +1/8 inch). The same three specimens shall then be subjected to condition B of the steam autoclave test. The autoclaved specimens shall then be retested for compression load deflection. The percent change in compression load deflection, based on the value before autoclaving, shall be determined.

4.3.12 Resistance to fluids. One specimen, a cube approximately 16 cubic centimeters (one cubic inch) in volume, shall be immersed for 48 ± 1 hour in each of the types of fluids specified in MIL-H-5606, MIL-T-5624, MIL-L-7808, MIL-L-23699 and MIL-H-83282. A different specimen shall be used for each fluid. Upon removal, the specimens shall be examined for visible evidence of softening or degradation. Each of the specimens shall comply with the requirements of 3.3.

4.3.13 Fungus resistance. Three specimens shall be prepared and tested in accordance with ASTM G 21. Specimens shall be 50 by 50 millimetre (2 by 2 inch) pieces as specified in ASTM G 21.

5. PACKAGING

Application. The requirements of section 5 apply only to purchase by or direct shipment to the Government.

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packing. Unless otherwise specified in the contract or purchase order (see 6.2), foams shall be packaged in quantities specified by the procuring activity in accordance with method III of MIL-P-116.

5.1.1.2 Intermediate packing. When required, specified quantities of unit packages shall be intermediately packaged as specified in the contract or purchase order (see 6.2).

5.1.2 Commercial. Foam preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

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5.2.1 Level A. Foam packaged as specified in 5.1 shall be separated by thickness and shall be packed in containers conforming to PPP-B-585 (class 3), or PPP-B-601 (overseas type). Unless otherwise specified (see 6.2) containers shall be provided with a case liner conforming to MIL-L-10547. Closure and strapping shall be in accordance with the appendix to the applicable container specifications. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds.

5.2.2 Level B. Foam packaged as specified in 5.1 shall be separated by thickness and shall be packed in containers conforming to PPP-B-636, grade V2s. Closure shall be in accordance with the appendix to the applicable container specification. The gross weight of the containers shall not exceed 150 pounds except for the PPP-B-636 box which shall not exceed the weight limitations of the box specification.

5.2.3 Commercial. Foam shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The flexible polyurethane foam is intended for use in aircraft under certain ballistic environments.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Foam dimensions and tolerances (see 3.5).

c. Periodic lot check test when required (see 4.2.3.3).

d. Request average flame spread index and range and average exposure time and range when testing in accordance with ASTM E 162 (see 4.3.4 and 4.3.5).

e. Selection of applicable levels of preservation and packing required including unit packing quantities and intermediate packing, if required (see 5.1 and 5.2).

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MIL-P-46111C(MR), Plastic Foam, Polyurethane (for Use in Aircraft)

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